NEET (UG) 2024

National Eligibility cum Entrance Test (Undergraduate)
Date: **05 May 2024** | Test Booklet Code: **T3**
Duration: 3 hours 20 minutes | 200 MCQs | Maximum Marks: 720

Important Instructions:

1. The Test Booklet contains **200** MCQs — 50 questions in each of **Physics, Chemistry, Botany, and Zoology**.
2. Each subject has two sections: **Section A** (35 questions, all compulsory) and **Section B** (15 questions; *attempt any 10* — first 10 attempted are evaluated).
3. Each correct response: **+4 marks**. Each incorrect response: –**1 mark**. Maximum marks: **720**.
4. Use a Blue/Black ballpoint pen only. Use of an electronic/manual calculator is prohibited.
5. The official answer key is provided at the end of this booklet.

PHYSICS (Questions 1–50)

SECTION A — Q1 to Q35 of the subject (all compulsory)

Q1. A tightly wound 100-turn coil of radius 10 cm carries a current of 7 A. The magnitude of the magnetic field at the centre of the coil is (μ■ = 4π × 10■■ SI units):

- **(1)** 4.4 mT
- **(2)** 44 T
- **(3)** 44 mT
- **(4)** 4.4 T

Q2. Match List-I (Material) with List-II (Susceptibility x): A. Diamagnetic B. Ferromagnetic C. Paramagnetic D. Non-magnetic ↔ I. x = 0 II. 0 > x ≥ –1 III. x ■ 1 IV. 0 < x < ε (small positive)

- **(1)** A-III, B-II, C-I, D-IV
- **(2)** A-IV, B-III, C-II, D-I
- **(3)** A-II, B-III, C-IV, D-I
- **(4)** A-II, B-I, C-III, D-IV

Q3. A thermodynamic system is taken through cycle abcda (rectangular P-V cycle: 100→400 cm³ at 100 kPa; 100→300 kPa at 400 cm³; 400→100 cm³ at 300 kPa; 300→100 kPa at 100 cm³). The work done by the gas along path bc is:

- **(1)** –90 J
- **(2)** –60 J
- **(3)** zero
- **(4)** 30 J

Q4. An unpolarised light beam strikes a glass surface at Brewster's angle. Then:

- **(1)** Both reflected and refracted light are completely polarised
- **(2)** Reflected light is completely polarised but refracted light is partially polarised
- **(3)** Reflected light will be partially polarised
- **(4)** Refracted light will be completely polarised

Q5. In an ideal transformer, turns ratio N■/N■ = 1/2. The ratio V■ : V■ is:

- **(1)** 1 : 1
- **(2)** 1 : 4
- **(3)** 1 : 2
- **(4)** 2 : 1

Q6. A logic circuit produces output Y as per the truth table: (A,B,Y) = (0,0,1),(0,1,0),(1,0,1),(1,1,0). Y equals:

- **(1)** B■
- **(2)** B
- **(3)** A·B + ■
- **(4)** A·B■ + ■

Q7. In a vernier calipers, (N+1) divisions of vernier coincide with N divisions of main scale. If 1 MSD = 0.1 mm, the vernier constant (in cm) is:

- **(1)** 100N
- **(2)** 10(N+1)
- **(3)** 1/(10N)
- **(4)** 1/[100(N+1)]

Q8. Maximum elongation of a 1 m steel wire (elastic limit 8×10■ N/m², Young's modulus 2×10^{11} N/m²) is:

(1) 40 mm
(2) 8 mm
(3) 4 mm
(4) 0.4 mm

Q9. A 10 N horizontal force is applied to block A (2 kg) which is in contact with block B (3 kg) on a frictionless surface. The force exerted by A on B is:

(1) 6 N
(2) 10 N
(3) zero
(4) 4 N

Q10. If the monochromatic source in Young's double-slit experiment is replaced by white light:

(1) Central bright white fringe surrounded by a few coloured fringes
(2) All bright fringes will be of equal width
(3) Interference pattern will disappear
(4) Central dark fringe surrounded by a few coloured fringes

Q11. Graph showing variation of $(1/\lambda^2)$ with kinetic energy E (λ is de Broglie wavelength of a free particle):

(1) Curve concave-up rising from origin
(2) Straight line through origin
(3) Curve concave-down rising from origin
(4) Vertical asymptote near zero E

Q12. In the given circuit (four 2 μF capacitors arranged with the bridge between A and B), the equivalent capacitance is:

(1) 0.5 μF
(2) 4 μF
(3) 2 μF
(4) 1 μF

Q13. A strong bar magnet is moved towards solenoid-2 from solenoid-1 (N pole faces solenoid 2). The induced currents in solenoid-1 (terminals A,B) and solenoid-2 (terminals C,D) are along:

(1) AB and CD
(2) BA and DC
(3) AB and DC
(4) BA and CD

Q14. Consider: A. For a solar cell, the I-V curve lies in the IV quadrant. B. In a reverse-biased pn diode, the μA current is due to majority charge carriers.

(1) Both A and B correct
(2) Both A and B incorrect
(3) A correct, B incorrect
(4) A incorrect, B correct

Q15. A light ray enters a right-angled prism at P with incidence 30°, travels parallel to base BC, and emerges along face AC. The refractive index of the prism is:

- **(1)** $\sqrt{3} / 4$
- **(2)** $\sqrt{3} / 2$
- **(3)** $\sqrt{5} / 4$
- **(4)** $\sqrt{5} / 2$

Q16. Assertion: Potential at axial point r = 2 m from centre of a dipole of moment 4 × 10■■ C·m is ±9 × 10^3 V. Reason: V = ±2P / (4πε■ r^2) for axial points.

- **(1)** A is true but R is false
- **(2)** A is false but R is true
- **(3)** Both A and R true; R is correct explanation
- **(4)** Both A and R true; R is NOT correct explanation

Q17. Moment of inertia of a thin rod about an axis through its midpoint perpendicular to it is 2400 g·cm^2. The length of the 400 g rod is nearly:

- **(1)** 20.7 cm
- **(2)** 72.0 cm
- **(3)** 8.5 cm
- **(4)** 17.5 cm

Q18. Terminal voltage of a battery (emf 10 V, internal resistance 1 Ω) connected to an external 4 Ω resistor is:

- **(1)** 8 V
- **(2)** 10 V
- **(3)** 4 V
- **(4)** 6 V

Q19. Match List-I (Spectral lines of H, n■→n■) with List-II (Wavelengths in nm): A. 3→2 B. 4→2 C. 5→2 D. 6→2 ↔ I. 410.2 II. 434.1 III. 656.3 IV. 486.1

- **(1)** A-IV, B-III, C-I, D-II
- **(2)** A-I, B-II, C-III, D-IV
- **(3)** A-II, B-I, C-IV, D-III
- **(4)** A-III, B-IV, C-II, D-I

Q20. Correct statements about a photon (c is speed of light): A. E = hν B. velocity = c C. p = hν/c D. In photon-electron collision both energy and momentum conserved E. Photon possesses positive charge

- **(1)** A, C and D only
- **(2)** A, B, D and E only
- **(3)** A and B only
- **(4)** A, B, C and D only

Q21. ²■■■■X →α→ Y →e■→ Z →β■→ P →e■→ Q. Mass and atomic numbers of Q are:

- **(1)** 288, 82
- **(2)** 286, 81
- **(3)** 280, 81
- **(4)** 286, 80

Q22. Displacement of a particle is $x = 2t - 1$ (SI units) under a 5 N force. Instantaneous power (SI units) is:

- **(1)** 7
- **(2)** 6
- **(3)** 10
- **(4)** 5

Q23. Output Y of the given logic gate (NAND of A; NOR of B; outputs into a NOR) is similar to a/an:

- **(1)** OR gate
- **(2)** AND gate
- **(3)** NAND gate
- **(4)** NOR gate

Q24. A planet has 1/10 the mass of Earth and half its diameter. Acceleration due to gravity on the planet is:

- **(1)** 4.9 m/s^2
- **(2)** 3.92 m/s^2
- **(3)** 19.6 m/s^2
- **(4)** 9.8 m/s^2

Q25. Statement I: Atoms are electrically neutral as they contain equal positive and negative charges. Statement II: Atoms of each element are stable and emit characteristic spectrum.

- **(1)** I correct, II incorrect
- **(2)** I incorrect, II correct
- **(3)** Both correct
- **(4)** Both incorrect

Q26. A wheel rolls on a level road with linear speed v. P is the highest point and Q is the lowest point on the wheel. Which is correct?

- **(1)** P and Q move with equal speeds
- **(2)** Point P has zero speed
- **(3)** Point P moves slower than Q
- **(4)** Point P moves faster than Q

Q27. A particle moving with uniform speed in a circular path maintains:

- **(1)** Constant velocity but varying acceleration
- **(2)** Varying velocity and varying acceleration
- **(3)** Constant velocity
- **(4)** Constant acceleration

Q28. A thin flat circular disc of radius 4.5 cm is placed gently on water (surface tension 0.07 N/m). Excess force required to lift it is:

- **(1)** 1.98 mN
- **(2)** 99 N
- **(3)** 19.8 mN
- **(4)** 198 N

Q29. In a uniform field of 0.049 T a magnetic needle ($I = 9.8 \times 10$[redacted] kg·m^2) makes 20 oscillations in 5 s. If $\mu = x \times 10$[redacted] A·m^2, then x equals:

- **(1)** $50\pi^2$
- **(2)** $1280\pi^2$
- **(3)** $5\pi^2$
- **(4)** $128\pi^2$

Q30. Two equal-mass bodies undergo a perfectly inelastic 1-D collision; A moves with v■, B is at rest. Velocity after collision is v■. v■ : v■ is:

- **(1)** 4 : 1
- **(2)** 1 : 4
- **(3)** 1 : 2
- **(4)** 2 : 1

Q31. $x = 5 \sin(\pi t + \pi/3)$ m represents SHM. Amplitude and time period are:

- **(1)** 5 cm, 1 s
- **(2)** 5 m, 1 s
- **(3)** 5 cm, 2 s
- **(4)** 5 m, 2 s

Q32. Quantities with the same dimensions as solid angle are:

- **(1)** Strain and arc
- **(2)** Angular speed and stress
- **(3)** Strain and angle
- **(4)** Stress and angle

Q33. A thin charged spherical shell (R = 3 cm, q = 1 μC). Potential difference between point C inside and point P on the surface (in V) is (1/4πε■ = 9 × 10■):

- **(1)** 0.5 × 10■
- **(2)** zero
- **(3)** 3 × 10■
- **(4)** 1 × 10■

Q34. A bob is whirled in a horizontal plane with initial speed ω rpm and string tension T. If the speed becomes 2ω at the same radius, the new tension is:

- **(1)** T/4
- **(2)** √2 T
- **(3)** T
- **(4)** 4T

Q35. A 100 Ω wire is divided into 10 equal parts. First 5 parts are connected in series, the next 5 in parallel; the two combinations are connected in series. The resistance of the final combination is:

- **(1)** 55 Ω
- **(2)** 60 Ω
- **(3)** 26 Ω
- **(4)** 52 Ω

SECTION B — Q36 to Q50 of the subject (attempt any 10)

Q36. T-V curves of an ideal gas at three pressures P■, P■, P■ shown alongside Charles's law (dotted). The order of pressures is:

- **(1)** P■ > P■ > P■
- **(2)** P■ > P■ > P■
- **(3)** P■ > P■ > P■
- **(4)** P■ > P■ > P■

Q37. A parallel plate capacitor is being charged through a resistor with current I. In the gap between the plates:

- **(1)** Displacement current of magnitude I flows opposite to I
- **(2)** Displacement current of magnitude greater than I flows in any direction
- **(3)** There is no current
- **(4)** Displacement current of magnitude I flows in the same direction as I

Q38. Which property is NOT true of an EM wave travelling in free space?

- **(1)** Travel with speed 1/√(μ■ε■)
- **(2)** Originate from charges moving with uniform speed
- **(3)** Transverse in nature
- **(4)** Energy density in E field equals energy density in B field

Q39. Which circuit can achieve the bridge balance (10 Ω, 10 Ω, 15 Ω, 5 Ω with galvanometer G and detector D)?

- **(1)** Configuration 1
- **(2)** Configuration 2
- **(3)** Configuration 3
- **(4)** Configuration 4

Q40. If the plates of a parallel plate capacitor connected to a battery are moved closer: A. charge stored increases B. energy stored decreases C. capacitance increases D. C = Q/V remains the same E. product of charge and voltage increases

- **(1)** B, D and E only
- **(2)** A, B and C only
- **(3)** A, B and E only
- **(4)** A, C and E only

Q41. A force $F = \alpha t^2 + \beta t$ acts on a particle. Which combination is dimensionless (α, β are constants)?

- **(1)** $\alpha\beta t$
- **(2)** $\alpha\beta/t$
- **(3)** $\beta t/\alpha$
- **(4)** $\alpha t/\beta$

Q42. A metallic bar ($Y = 0.5 \times 10^{11}$ N/m², α = 10■■ /°C, L = 1 m, A = 10■³ m²) is heated from 0 to 100 °C without expansion. Compressive force is:

- **(1)** 100×10^3 N
- **(2)** 2×10^3 N
- **(3)** 52×10^3 N
- **(4)** 50×10^3 N

Q43. A small telescope: objective f = 140 cm, eyepiece f = 5.0 cm. Magnifying power for a distant object is:

- **(1)** 17
- **(2)** 32
- **(3)** 34
- **(4)** 28

Q44. An iron bar of length L has magnetic moment M. Bent at midpoint so the two arms make 60° with each other. New magnetic moment is:

- **(1)** 2M
- **(2)** M/√3
- **(3)** M
- **(4)** M/2

Q45. A 10 μF capacitor is connected to 210 V, 50 Hz source. Peak current (π = 3.14) is nearly:

- **(1)** 1.20 A
- **(2)** 0.35 A
- **(3)** 0.58 A
- **(4)** 0.93 A

Q46. Two heaters rated 1 kW and 2 kW are connected first in series and then in parallel to a fixed source. Ratio of total powers (series : parallel) is:

- **(1)** 1 : 2
- **(2)** 2 : 3
- **(3)** 1 : 1
- **(4)** 2 : 9

Q47. From the v-t plot (rises linearly, plateau at constant v, then falls linearly to zero), which a-t graph best fits?

- **(1)** Positive pulse, zero, negative pulse
- **(2)** Positive ramp, zero, negative ramp
- **(3)** Trapezoidal pulse
- **(4)** Step function rising

Q48. A pendulum's bob mass is increased to 3× and length is halved. New time period is (x/2) × original. Then x is:

- **(1)** 2√3
- **(2)** 4
- **(3)** √3
- **(4)** √2

Q49. Minimum energy to launch a satellite of mass m from Earth (mass M, radius R) into a circular orbit at altitude 2R is:

- **(1)** GMm/(2R)
- **(2)** GMm/(3R)
- **(3)** 5GMm/(6R)
- **(4)** 2GMm/(3R)

Q50. A sheet is placed in front of a strong magnetic pole. A force is needed to: A. hold it if magnetic B. hold it if non-magnetic C. move it away with uniform velocity if conducting D. move it away with uniform velocity if non-conducting and non-polar

- **(1)** A, C and D only
- **(2)** C only
- **(3)** B and D only
- **(4)** A and C only

CHEMISTRY (Questions 51–100)

SECTION A — Q1 to Q35 of the subject (all compulsory)

Q51. Match (Conversion) ↔ (Faraday required): A. 1 mol H■O → O■ B. 1 mol MnO■■ → Mn^{2}■ C. 1.5 mol Ca from molten CaCl■ D. 1 mol FeO → Fe■O■ ↔ I. 3F II. 2F III. 1F IV. 5F

- **(1)** A-II, B-III, C-I, D-IV
- **(2)** A-III, B-IV, C-II, D-I
- **(3)** A-II, B-IV, C-I, D-III
- **(4)** A-III, B-IV, C-I, D-II

Q52. Which reaction is NOT a redox reaction?

- **(1)** H■ + Cl■ → 2 HCl
- **(2)** BaCl■ + Na■SO■ → BaSO■ + 2 NaCl
- **(3)** Zn + CuSO■ → ZnSO■ + Cu
- **(4)** 2 KClO■ + I■ → 2 KIO■ + Cl■

Q53. Intramolecular hydrogen bonding is present in:

- **(1)** m-Nitrophenol
- **(2)** HF
- **(3)** o-Nitrophenol
- **(4)** p-Nitrophenol

Q54. Fehling's solution 'A' is:

- **(1)** Alkaline solution of sodium potassium tartrate (Rochelle's salt)
- **(2)** Aqueous sodium citrate
- **(3)** Aqueous copper sulphate
- **(4)** Alkaline copper sulphate

Q55. 1 g of NaOH is treated with 25 mL of 0.75 M HCl. Mass of NaOH unreacted:

- **(1)** Zero mg
- **(2)** 200 mg
- **(3)** 750 mg
- **(4)** 250 mg

Q56. Match (Compound) ↔ (Shape/Geometry): A. NH■ B. BrF■ C. XeF■ D. SF■ ↔ I. Trigonal pyramidal II. Square planar III. Octahedral IV. Square pyramidal

- **(1)** A-III, B-IV, C-I, D-II
- **(2)** A-II, B-III, C-IV, D-I
- **(3)** A-I, B-IV, C-II, D-III
- **(4)** A-II, B-IV, C-III, D-I

Q57. E°(Mn^{3}■/Mn^{2}■) is more positive than E°(Cr^{3}■/Cr^{2}■) and E°(Fe^{3}■/Fe^{2}■) because of change of:

- **(1)** d■ to d■
- **(2)** d^{3} to d■
- **(3)** d■ to d■
- **(4)** d■ to d^{2}

Q58. Match (Process) ↔ (Conditions): A. Isothermal B. Isochoric C. Isobaric D. Adiabatic ↔ I. No heat exchange II. Constant T III. Constant V IV. Constant P

- **(1)** A-I, B-II, C-III, D-IV
- **(2)** A-II, B-III, C-IV, D-I
- **(3)** A-IV, B-III, C-II, D-I
- **(4)** A-IV, B-II, C-III, D-I

Q59. Activation energy of a reaction can be calculated if one knows:

- **(1)** Orientation of reactant molecules
- **(2)** Rate constants at two different temperatures
- **(3)** Rate constant at standard temperature
- **(4)** Probability of collision

Q60. A C■H■■ compound has two tertiary carbons. Its IUPAC name is:

- **(1)** 2,3-dimethylbutane
- **(2)** 2,2-dimethylbutane
- **(3)** n-hexane
- **(4)** 2-methylpentane

Q61. 'Spin only' magnetic moment is the same for which ions? A. Ti^{3}■ B. Cr^{2}■ C. Mn^{2}■ D. Fe^{2}■ E. Sc^{3}■

- **(1)** B and C only
- **(2)** A and D only
- **(3)** B and D only
- **(4)** A and E only

Q62. Increasing electronegativity of N, O, F, C, Si:

- **(1)** O < F < N < C < Si
- **(2)** F < O < N < C < Si
- **(3)** Si < C < N < O < F
- **(4)** Si < C < O < N < F

Q63. Which alcohol reacts instantaneously with Lucas reagent?

- **(1)** (CH■)■CH-CH■OH (isobutyl)
- **(2)** (CH■)■C-OH (tert-butyl)
- **(3)** n-Butanol
- **(4)** sec-Butanol

Q64. Statement I: [Co(NH■)■]³■ and [CoF■]³■ are both octahedral but differ in magnetic behaviour. Statement II: [Co(NH■)■]³■ is diamagnetic; [CoF■]³■ is paramagnetic.

- **(1)** I true, II false
- **(2)** I false, II true
- **(3)** Both true
- **(4)** Both false

Q65. Statement I: BP of Group 16 hydrides: H■O > H■Te > H■Se > H■S. Statement II: H■O has higher BP than expected because of extensive H-bonding.

- **(1)** I true, II false
- **(2)** I false, II true
- **(3)** Both true
- **(4)** Both false

Q66. Match (Quantum Number) ↔ (Information): A. m■ B. m■ C. l D. n ↔ I. shape of orbital II. size of orbital III. orientation of orbital IV. orientation of spin

- **(1)** A-III, B-IV, C-II, D-I
- **(2)** A-II, B-I, C-IV, D-III
- **(3)** A-I, B-III, C-II, D-IV
- **(4)** A-III, B-IV, C-I, D-II

Q67. Match (Reaction) ↔ (Reagents/Condition): A. Biphenyl → cyclohexanone (one C=O) B. Benzene → benzophenone C. Cyclohexanol → cyclohexanone D. Ethylbenzene → benzoate ↔ I. PhCOCl/AlCl■ II. CrO■ III. KMnO■/KOH, Δ IV. (i) O■ (ii) Zn-H■O

- **(1)** A-IV, B-I, C-II, D-III
- **(2)** A-I, B-IV, C-II, D-III
- **(3)** A-IV, B-I, C-III, D-II
- **(4)** A-III, B-I, C-II, D-IV

Q68. Reagents to convert PhCH■-CH=CH■ → PhCH■-CH■-CHO:

- **(1)** (i) BH■ (ii) H■O■/OH■ (iii) alk. KMnO■ (iv) H■O■
- **(2)** (i) H■O/H■ (ii) PCC
- **(3)** (i) H■O/H■ (ii) CrO■
- **(4)** (i) BH■ (ii) H■O■/OH■ (iii) PCC

Q69. Reagents with which glucose does NOT react: A. Tollen's B. Schiff's C. HCN D. NH■OH E. NaHSO■

- **(1)** B and E
- **(2)** E and D
- **(3)** B and C
- **(4)** A and D

Q70. Match (Molecule) ↔ (Bonds between two C atoms): A. Ethane B. Ethene C. C■ D. Ethyne ↔ I. one σ and two π II. two π III. one σ IV. one σ and one π

- **(1)** A-III, B-IV, C-II, D-I
- **(2)** A-III, B-IV, C-I, D-II
- **(3)** A-I, B-IV, C-II, D-III
- **(4)** A-IV, B-III, C-II, D-I

Q71. Among Group 16 elements, which does NOT show –2 oxidation state?

- **(1)** Te
- **(2)** Po
- **(3)** O
- **(4)** Se

Q72. For 2A ■ B + C, Kc = 4 × 10■³. At a moment [A] = [B] = [C] = 2 × 10■³ M. Then:

- **(1)** Reaction tends backward
- **(2)** Reaction has gone to completion forward
- **(3)** Reaction is at equilibrium
- **(4)** Reaction tends forward

Q73. Which plot of ln k vs 1/T is consistent with the Arrhenius equation?

- **(1)** Increasing line
- **(2)** Decreasing line
- **(3)** Increasing line through origin
- **(4)** Decreasing line through origin

Q74. In which equilibrium are K■ and K_c NOT equal?

- **(1)** CO + H■O ■ CO■ + H■
- **(2)** 2 BrCl ■ Br■ + Cl■
- **(3)** PCl■ ■ PCl■ + Cl■
- **(4)** H■ + I■ ■ 2 HI

Q75. Statement I: BP of three pentanes: n-pentane > isopentane > neopentane. Statement II: With more branching the molecule becomes more spherical, surface area decreases, intermolecular forces weaken, BP decreases.

- **(1)** I correct, II incorrect
- **(2)** I incorrect, II correct
- **(3)** Both correct
- **(4)** Both incorrect

Q76. Compound that undergoes S■¹ reaction the fastest:

- **(1)** PhBr (bromobenzene)
- **(2)** PhCH(CH■)Br (1-phenylethyl bromide)
- **(3)** Cyclohexyl-CH■Br (primary)
- **(4)** Cyclohexyl-Br (secondary)

Q77. Energy of an electron in n=1 of He■ is –x J. Energy of an electron in n=2 of Be³■ (in J) is:

- **(1)** –4x
- **(2)** –4x/9
- **(3)** –x
- **(4)** –x/9

Q78. Entropy increases in: A. liquid → vapour B. crystalline solid 130 K → 0 K C. 2 NaHCO■ → Na■CO■ + CO■ + H■O D. Cl■ → 2 Cl

- **(1)** A, C and D
- **(2)** C and D
- **(3)** A and C
- **(4)** A, B and D

Q79. Solid → vapour without passing through liquid; the purification technique used is:

- **(1)** Distillation
- **(2)** Chromatography
- **(3)** Crystallization
- **(4)** Sublimation

Q80. Match (Complex) ↔ (Type of isomerism): A. [Co(NH■)■(NO■)]Cl■ B. [Co(NH■)■(SO■)]Br C. [Co(NH■)■][Cr(CN)■] D. [Co(H■O)■]Cl■ ↔ I. Solvate II. Linkage III. Ionization IV. Coordination

- **(1)** A-I, B-IV, C-III, D-II
- **(2)** A-II, B-IV, C-III, D-I
- **(3)** A-II, B-III, C-IV, D-I
- **(4)** A-I, B-III, C-IV, D-II

Q81. Statement I: Aniline does not undergo Friedel-Crafts alkylation. Statement II: Aniline cannot be prepared through Gabriel synthesis.

- **(1)** I correct, II false
- **(2)** I incorrect, II true
- **(3)** Both true
- **(4)** Both false

Q82. Increasing first IE of Li, Be, B, C, N:

- **(1)** Li < Be < C < B < N
- **(2)** Li < Be < N < B < C
- **(3)** Li < Be < B < C < N
- **(4)** Li < B < Be < C < N

Q83. Highest number of helium atoms is in:

- **(1)** 4 g of helium
- **(2)** 2.271098 L of helium at STP
- **(3)** 4 mol of helium
- **(4)** 4 u of helium

Q84. The most stable carbocation among the following is:

- **(1)** Cyclopentyl-CH■■ (primary)
- **(2)** 1-methylcyclohexyl■ (3° on ring)
- **(3)** (CH■)■CH-CH■-CH■ (secondary, branched)
- **(4)** (CH■)CH■CH(CH■)CH■(CH■) (secondary)

Q85. Henry's law constants for gases A, B, C in water are 145, 2 × 10■■ and 35 kbar. Solubility order:

- **(1)** A > C > B
- **(2)** A > B > C
- **(3)** B > A > C
- **(4)** B > C > A

SECTION B — Q36 to Q50 of the subject (attempt any 10)

Q86. A compound X contains 32% A, 20% B, rest C. Empirical formula (atomic masses A=64, B=40, C=32):

- **(1)** AB■C■
- **(2)** ABC■
- **(3)** A■BC■
- **(4)** ABC■

Q87. Products A and B in: 3 ROH + PCl■ → 3 RCl + A; ROH + PCl■ → RCl + HCl + B

- **(1)** H■PO■ and POCl■
- **(2)** H■PO■ and POCl■
- **(3)** POCl■ and H■PO■
- **(4)** POCl■ and H■PO■

Q88. Plot of osmotic pressure vs concentration (mol/L) is a straight line with slope 25.73 L·bar/mol. The temperature (R = 0.083 L·bar/mol·K) is:

- **(1)** 25.73 °C
- **(2)** 12.05 °C
- **(3)** 37 °C
- **(4)** 310 °C

Q89. PhCH=CHPh + KMnO■/H■ → P (major). P is:

- **(1)** Ph-CH(OH)-CH(OH)-Ph (diol)
- **(2)** Ph-CO-CO-Ph (diketone)
- **(3)** PhCHO (benzaldehyde)
- **(4)** PhCOOH (benzoic acid)

Q90. Statement I: [Co(NH■)■]^{3}■ is homoleptic; [Co(NH■)■Cl■]■ is heteroleptic. Statement II: [Co(NH■)■]^{3}■ has only one kind of ligand; [Co(NH■)■Cl■]■ has more than one kind.

- **(1)** I true, II false
- **(2)** I false, II true
- **(3)** Both true
- **(4)** Both false

Q91. Acid added during preparation of Mohr's salt to prevent hydrolysis of Fe^{2}■:

- **(1)** Dilute nitric acid
- **(2)** Dilute sulphuric acid
- **(3)** Dilute hydrochloric acid
- **(4)** Concentrated sulphuric acid

Q92. Identify the correct answer:

- **(1)** Dipole moment of NF■ > NH■
- **(2)** Three canonical forms can be drawn for CO■^{2}■
- **(3)** Three resonance structures can be drawn for ozone
- **(4)** BF■ has non-zero dipole moment

Q93. Increasing group number (0 → VI) for cations: A. Al^{3}■ B. Cu^{2}■ C. Ba^{2}■ D. Co^{2}■ E. Mg^{2}■

- **(1)** E, C, D, B, A
- **(2)** E, A, B, C, D
- **(3)** B, A, D, C, E
- **(4)** B, C, A, D, E

Q94. CH■CH■CH■I →NaCN→ A →OH■ partial hydrolysis→ B →NaOH/Br■→ C (major). C is:

- **(1)** Butanamide
- **(2)** α-bromobutanoic acid
- **(3)** Propylamine
- **(4)** Butylamine

Q95. Rate quadruples when T changes from 27 °C to 57 °C. Activation energy (R = 8.314, log 4 = 0.6021):

- **(1)** 3.80 kJ/mol
- **(2)** 3804 kJ/mol
- **(3)** 38.04 kJ/mol
- **(4)** 380.4 kJ/mol

Q96. At equilibrium [N■] = 3.0×10■^{3}, [O■] = 4.2×10■^{3}, [NO] = 2.8×10■^{3} M for 2 NO ■ N■ + O■. If 0.1 M NO is taken in a closed vessel, degree of dissociation α at equilibrium:

- **(1)** 0.8889
- **(2)** 0.717
- **(3)** 0.00889
- **(4)** 0.0889

Q97. Work done in reversible isothermal expansion of 1 mol H■ at 25 °C from 20 atm to 10 atm (R = 2 cal/K/mol):

- **(1)** 413.14 cal
- **(2)** 100 cal
- **(3)** 0 cal
- **(4)** –413.14 cal

Q98. Mass of Cu deposited by 9.6487 A through CuSO■ for 100 s (M = 63 g/mol, F = 96487 C):

- **(1)** 31.5 g
- **(2)** 0.0315 g
- **(3)** 3.15 g
- **(4)** 0.315 g

Q99. 2-methylcyclohexan-1-ol → PBr■ → A → alc. KOH/Δ → B (major). A and B are:

- **(1)** A: 1-Br, 2-OH cyclohexane; B: 1-OH cyclohex-2-ene
- **(2)** A: 1-Br,2-OH cyclohexane; B: 2-methyl-cyclohex-1-one
- **(3)** A: 1-Br,2-Me cyclohexane; B: 1-methyl cyclohexene
- **(4)** A: 1-Br,2-Me cyclohexane; B: 3-methyl cyclohexene

Q100. The pair of lanthanoid ions which are diamagnetic:

- **(1)** Gd^{3}■ and Eu^{3}■
- **(2)** Pm^{3}■ and Sm^{3}■
- **(3)** Ce■■ and Yb^{2}■
- **(4)** Ce^{3}■ and Eu^{2}■

BOTANY (Questions 101–150)

SECTION A — Q1 to Q35 of the subject (all compulsory)

Q101. Identify the set of correct statements: A. Vallisneria flowers are colourful and produce nectar B. Waterlily is not pollinated by water C. In most water-pollinated species, pollen grains are protected from wetting D. Pollen grains of some hydrophytes are long and ribbon-like E. In some hydrophytes, pollen grains are carried passively in water

- **(1)** A, C, D and E only
- **(2)** B, C, D and E only
- **(3)** C, D and E only
- **(4)** A, B, C and D only

Q102. Conservation in which threatened species are removed from natural habitat to a special protected setting is called:

- **(1)** Semi-conservative method
- **(2)** Sustainable development
- **(3)** in-situ conservation
- **(4)** Biodiversity conservation

Q103. Inhibition of succinic dehydrogenase by malonate is a classical example of:

- **(1)** Competitive inhibition
- **(2)** Enzyme activation
- **(3)** Cofactor inhibition
- **(4)** Feedback inhibition

Q104. Identify the part of the seed (labelled A, B, C, D) destined to form the root upon germination:

- **(1)** C
- **(2)** D
- **(3)** A
- **(4)** B

Q105. Bulliform cells are responsible for:

- **(1)** Increased photosynthesis in monocots
- **(2)** Providing large spaces for storage of sugars
- **(3)** Inward curling of leaves in monocots
- **(4)** Protecting the plant from salt stress

Q106. Required for the dark reaction of photosynthesis: A. Light B. Chlorophyll C. CO■ D. ATP E. NADPH

- **(1)** C, D and E only
- **(2)** D and E only
- **(3)** A, B and C only
- **(4)** B, C and D only

Q107. Formation of interfascicular cambium from fully developed parenchyma cells is an example of:

- **(1)** Dedifferentiation
- **(2)** Maturation
- **(3)** Differentiation
- **(4)** Redifferentiation

Q108. Hind II always cuts DNA at a recognition sequence consisting of:

- **(1)** 4 bp
- **(2)** 10 bp
- **(3)** 8 bp
- **(4)** 6 bp

Q109. Tropical regions show greatest species richness because: A. relatively undisturbed for millions of years B. tropical environments are more seasonal C. more solar energy in tropics D. constant environments promote niche specialization E. tropical environments are constant and predictable

- **(1)** A, B and E only
- **(2)** A, B and D only
- **(3)** A, C, D and E only
- **(4)** A and B only

Q110. Which is NOT a criterion for classification of fungi?

- **(1)** Mode of spore formation
- **(2)** Fruiting body
- **(3)** Morphology of mycelium
- **(4)** Mode of nutrition

Q111. ATP and NADPH required per CO■ fixed in Calvin cycle:

- **(1)** 3 ATP and 3 NADPH
- **(2)** 3 ATP and 2 NADPH
- **(3)** 2 ATP and 3 NADPH
- **(4)** 2 ATP and 2 NADPH

Q112. Major causes of biodiversity loss: A. Over-exploitation B. Co-extinction C. Mutation D. Habitat loss and fragmentation E. Migration

- **(1)** A, B and E only
- **(2)** A, B and D only
- **(3)** A, C and D only
- **(4)** A, B, C and D only

Q113. The capacity to generate a whole plant from any cell is called:

- **(1)** Differentiation
- **(2)** Somatic hybridization
- **(3)** Totipotency
- **(4)** Micropropagation

Q114. In Verhulst-Pearl logistic growth $dN/dt = rN[(K–N)/K]$, K is:

- **(1)** Carrying capacity
- **(2)** Population density
- **(3)** Intrinsic rate of natural increase
- **(4)** Biotic potential

Q115. Spindle fibres attach to kinetochores during:

- **(1)** Anaphase
- **(2)** Telophase
- **(3)** Prophase
- **(4)** Metaphase

Q116. Identify the type of flowers based on position of calyx, corolla, androecium w.r.t. ovary in figures (a) and (b):

- **(1)** (a) Perigynous; (b) Epigynous
- **(2)** (a) Perigynous; (b) Perigynous
- **(3)** (a) Epigynous; (b) Hypogynous
- **(4)** (a) Hypogynous; (b) Epigynous

Q117. Match: A. Rhizopus B. Ustilago C. Puccinia D. Agaricus ↔ I. Mushroom II. Smut fungus III. Bread mould IV. Rust fungus

- **(1)** A-III, B-II, C-I, D-IV
- **(2)** A-IV, B-III, C-II, D-I
- **(3)** A-III, B-II, C-IV, D-I
- **(4)** A-I, B-III, C-II, D-IV

Q118. Black seed colour (BB/Bb) is dominant over white (bb). To find the genotype of a black-seed plant, cross with:

- **(1)** Bb
- **(2)** BB/Bb
- **(3)** BB
- **(4)** bb

Q119. A pink-flowered Snapdragon × red-flowered Snapdragon. Phenotypes in progeny:

- **(1)** Only pink
- **(2)** Red, pink and white
- **(3)** Only red
- **(4)** Red and pink only

Q120. Match: A. Two or more alternative forms of a gene B. Cross of F■ with homozygous recessive parent C. Cross of F■ with any of the parents D. Number of chromosome sets ↔ I. Back cross II. Ploidy III. Allele IV. Test cross

- **(1)** A-III, B-IV, C-I, D-II
- **(2)** A-IV, B-III, C-II, D-I
- **(3)** A-I, B-II, C-III, D-IV
- **(4)** A-II, B-I, C-III, D-IV

Q121. Lecithin (a small molecular-weight organic compound in living tissues) is an example of:

- **(1)** Glycerides
- **(2)** Carbohydrates
- **(3)** Amino acids
- **(4)** Phospholipids

Q122. Match: A. Clostridium butylicum B. Saccharomyces cerevisiae C. Trichoderma polysporum D. Streptococcus sp. ↔ I. Ethanol II. Streptokinase III. Butyric acid IV. Cyclosporin-A

- **(1)** A-III, B-I, C-IV, D-II
- **(2)** A-IV, B-I, C-III, D-II
- **(3)** A-III, B-I, C-II, D-IV
- **(4)** A-II, B-IV, C-III, D-I

Q123. In the given figure (guard cells of stomata), which component has thin outer walls and highly thickened inner walls?

(1) A
(2) B
(3) C
(4) D

Q124. Example of an actinomorphic flower:

(1) Pisum
(2) Sesbania
(3) Datura
(4) Cassia

Q125. A transcription unit in DNA has three regions (upstream $\rightarrow$ downstream):

(1) Inducer, Repressor, Structural gene
(2) Promoter, Structural gene, Terminator
(3) Repressor, Operator, Structural gene
(4) Structural gene, Transposons, Operator

Q126. Fate of a piece of DNA carrying only a gene of interest transferred into an alien organism: A. multiplies independently in progeny B. may integrate into recipient genome C. may multiply and be inherited along with host DNA D. it is not an integrated part of chromosome E. ability to replicate

(1) B and C only
(2) A and E only
(3) A and B only
(4) D and E only

Q127. Auxin is used by gardeners to weed lawns without harming grass because auxin:

(1) Does not affect mature monocotyledonous plants
(2) Helps in cell division in grasses
(3) Promotes apical dominance
(4) Promotes abscission of mature leaves only

Q128. Cofactor of carboxypeptidase:

(1) Flavin
(2) Haem
(3) Zinc
(4) Niacin

Q129. Lactose is transported into bacterial cells by the action of:

(1) Permease
(2) Polymerase
(3) β-galactosidase
(4) Acetylase

Q130. Mendel's Law of Dominance explains: A. one of a pair of factors is dominant, the other recessive B. alleles do not show any expression and both characters appear in F■ C. factors occur in pairs in normal diploids D. the discrete unit controlling a character is called factor E. expression of only one parental character in monohybrid cross

- **(1)** B, C and D only
- **(2)** A, B, C, D and E
- **(3)** A, B and C only
- **(4)** A, C, D and E only

Q131. Statement I: Bt toxins are insect-group specific and coded by the gene cry IAc. Statement II: Bt toxin exists as inactive protoxin in B. thuringiensis; after ingestion, the acidic pH of the insect gut activates it.

- **(1)** I true, II false
- **(2)** I false, II true
- **(3)** Both true
- **(4)** Both false

Q132. Statement I: Parenchyma is living but collenchyma is dead. Statement II: Gymnosperms lack xylem vessels; presence of xylem vessels is characteristic of angiosperms.

- **(1)** I true, II false
- **(2)** I false, II true
- **(3)** Both true
- **(4)** Both false

Q133. Statement I: Chromosomes become gradually visible under light microscope during leptotene. Statement II: Diplotene begins with dissolution of synaptonemal complex.

- **(1)** I true, II false
- **(2)** I false, II true
- **(3)** Both true
- **(4)** Both false

Q134. Match: A. Nucleolus B. Centriole C. Leucoplasts D. Golgi apparatus ↔ I. Site of formation of glycolipid II. Cartwheel organisation III. Site for ribosomal RNA synthesis IV. Storing nutrients

- **(1)** A-III, B-IV, C-II, D-I
- **(2)** A-I, B-II, C-III, D-IV
- **(3)** A-III, B-II, C-IV, D-I
- **(4)** A-II, B-III, C-I, D-IV

Q135. List of endangered species was released by:

- **(1)** FOAM
- **(2)** IUCN
- **(3)** GEAC
- **(4)** WWF

SECTION B — Q36 to Q50 of the subject (attempt any 10)

Q136. DNA in chloroplast is:

- **(1)** Linear, single stranded
- **(2)** Circular, single stranded
- **(3)** Linear, double stranded
- **(4)** Circular, double stranded

Q137. What are fused in somatic hybridization between two plant varieties?

- **(1)** Protoplasts
- **(2)** Pollens
- **(3)** Callus
- **(4)** Somatic embryos

Q138. Identify the description of the figure (compact spike-like inflorescence with prominent exposed stamens):

- **(1)** Cleistogamous flowers showing autogamy
- **(2)** Compact inflorescence showing complete autogamy
- **(3)** Wind-pollinated plant inflorescence with well-exposed stamens
- **(4)** Water-pollinated flowers with mucilaginous covering

Q139. Spraying sugarcane with which PGR increases stem length and yield?

- **(1)** Cytokinin
- **(2)** Abscisic acid
- **(3)** Auxin
- **(4)** Gibberellin

Q140. Match: A. Frederick Griffith B. Jacob & Monod C. Khorana D. Meselson & Stahl ↔ I. Genetic code II. Semi-conservative DNA replication III. Transformation IV. Lac operon

- **(1)** A-II, B-III, C-IV, D-I
- **(2)** A-IV, B-I, C-II, D-III
- **(3)** A-III, B-II, C-I, D-IV
- **(4)** A-III, B-IV, C-I, D-II

Q141. Match: A. GLUT-4 B. Insulin C. Trypsin D. Collagen ↔ I. Hormone II. Enzyme III. Intercellular ground substance IV. Glucose transporter

- **(1)** A-II, B-III, C-IV, D-I
- **(2)** A-III, B-IV, C-I, D-II
- **(3)** A-IV, B-I, C-II, D-III
- **(4)** A-I, B-II, C-III, D-IV

Q142. Statement I: In C■ plants, some O■ binds RuBisCO; CO■ fixation is decreased. Statement II: In C■ plants, mesophyll cells show very little photorespiration; bundle sheath cells show none.

- **(1)** I true, II false
- **(2)** I false, II true
- **(3)** Both true
- **(4)** Both false

Q143. Step in TCA cycle that does NOT involve oxidation of substrate:

- **(1)** Succinyl-CoA → Succinic acid
- **(2)** Isocitrate → α-ketoglutaric acid
- **(3)** Malic acid → Oxaloacetic acid
- **(4)** Succinic acid → Malic acid

Q144. Match: A. Citric acid cycle B. Glycolysis C. Electron transport system D. Proton gradient ↔ I. Cytoplasm II. Mitochondrial matrix III. Intermembrane space IV. Inner mitochondrial membrane

- **(1)** A-III, B-IV, C-I, D-II
- **(2)** A-IV, B-III, C-II, D-I
- **(3)** A-I, B-II, C-III, D-IV
- **(4)** A-II, B-I, C-IV, D-III

Q145. Correct statement on E. coli replication:

- **(1)** DNA-dependent DNA polymerase polymerises 5'→3' AND 3'→5'
- **(2)** DNA-dependent DNA polymerase polymerises only 5'→3'
- **(3)** DNA-dependent DNA polymerase polymerises only 3'→5'
- **(4)** DNA-dependent RNA polymerase polymerises only 5'→3'

Q146. If NPP of first trophic level = 100x kcal/m²/yr, GPP of third trophic level (assuming 10% rule) is:

- **(1)** 10x kcal/m²/yr
- **(2)** 100x/3x kcal/m²/yr
- **(3)** x/10 kcal/m²/yr
- **(4)** x kcal/m²/yr

Q147. Match: A. Rose B. Pea C. Cotton D. Mango ↔ I. Twisted aestivation II. Perigynous flower III. Drupe IV. Marginal placentation

- **(1)** A-IV, B-III, C-II, D-I
- **(2)** A-II, B-III, C-IV, D-I
- **(3)** A-II, B-IV, C-I, D-III
- **(4)** A-I, B-II, C-III, D-IV

Q148. Match: A. Robert May B. Alexander von Humboldt C. Paul Ehrlich D. David Tilman ↔ I. Species-area relationship II. Long-term ecosystem outdoor plot experiment III. Global species diversity ≈ 7 million IV. Rivet popper hypothesis

- **(1)** A-I, B-III, C-II, D-IV
- **(2)** A-III, B-IV, C-II, D-I
- **(3)** A-II, B-III, C-I, D-IV
- **(4)** A-III, B-I, C-IV, D-II

Q149. Match (Stamen type) ↔ (Example): A. Monoadelphous B. Diadelphous C. Polyadelphous D. Epiphyllous ↔ I. Citrus II. Pea III. Lily IV. China-rose

- **(1)** A-I, B-II, C-IV, D-III
- **(2)** A-III, B-I, C-IV, D-II
- **(3)** A-IV, B-II, C-I, D-III
- **(4)** A-IV, B-I, C-II, D-III

Q150. Correct statements about Phaeophyceae: A. asexual by biflagellate zoospores B. sexual is oogamous only C. stored food is mannitol or laminarin D. major pigments: chl a, c, carotenoids, xanthophyll E. cellulosic wall covered by gelatinous algin

- **(1)** A, C, D and E only
- **(2)** A, B, C and E only
- **(3)** A, B, C and D only
- **(4)** B, C, D and E only

ZOOLOGY (Questions 151–200)

SECTION A — Q1 to Q35 of the subject (all compulsory)

Q151. Match: A. Typhoid B. Leishmaniasis C. Ringworm D. Filariasis ↔ I. Fungus II. Nematode III. Protozoa IV. Bacteria

- **(1)** A-III, B-I, C-IV, D-II
- **(2)** A-II, B-IV, C-III, D-I
- **(3)** A-I, B-III, C-II, D-IV
- **(4)** A-IV, B-III, C-I, D-II

Q152. Match: A. Non-medicated IUD B. Copper-releasing IUD C. Hormone-releasing IUD D. Implants ↔ I. Multiload 375 II. Progestogens III. Lippes loop IV. LNG-20

- **(1)** A-IV, B-I, C-II, D-III
- **(2)** A-III, B-I, C-IV, D-II
- **(3)** A-III, B-I, C-II, D-IV
- **(4)** A-I, B-III, C-IV, D-II

Q153. Statement I: Presence/absence of hymen is not a reliable indicator of virginity. Statement II: The hymen is torn during the first coitus only.

- **(1)** I true, II false
- **(2)** I false, II true
- **(3)** Both true
- **(4)** Both false

Q154. In both sexes of cockroach, anal cerci are present on:

- **(1)** 8th and 9th segment
- **(2)** 11th segment
- **(3)** 5th segment
- **(4)** 10th segment

Q155. Match: A. Pons B. Hypothalamus C. Medulla D. Cerebellum ↔ I. Provides additional space for neurons; regulates posture and balance II. Controls respiration and gastric secretions III. Connects different brain regions IV. Neurosecretory cells

- **(1)** A-I, B-III, C-II, D-IV
- **(2)** A-II, B-I, C-III, D-IV
- **(3)** A-II, B-III, C-I, D-IV
- **(4)** A-III, B-IV, C-II, D-I

Q156. Which is NOT a steroid hormone?

- **(1)** Progesterone
- **(2)** Glucagon
- **(3)** Cortisol
- **(4)** Testosterone

Q157. DNA-dependent RNA polymerase product for template 3' TACATGGCAAATATCCATTCA 5':

- **(1)** 5' AUGUACCGUUUAUAGGGAAGU 3'
- **(2)** 5' ATGTACCGTTTATAGGTAAGT 3'
- **(3)** 5' AUGUACCGUUUAUAGGUAAGU 3'
- **(4)** 5' AUGUAAAGUUUAUAGGUAAGU 3'

Q158. Three types of muscles (a, b, c) — identify the matching pair with locations:

- **(1)** (a) Skeletal-Biceps; (b) Involuntary-Intestine; (c) Smooth-Heart
- **(2)** (a) Involuntary-Nose tip; (b) Skeletal-Bone; (c) Cardiac-Heart
- **(3)** (a) Smooth-Toes; (b) Skeletal-Legs; (c) Cardiac-Heart
- **(4)** (a) Skeletal-Triceps; (b) Smooth-Stomach; (c) Cardiac-Heart

Q159. Stages of cell division: A. G2 B. Cytokinesis C. S phase D. Karyokinesis E. G1. Correct sequence:

- **(1)** B-D-E-A-C
- **(2)** E-C-A-D-B
- **(3)** C-E-D-A-B
- **(4)** E-B-D-A-C

Q160. Autoimmune disorders: A. Myasthenia gravis B. Rheumatoid arthritis C. Gout D. Muscular dystrophy E. SLE

- **(1)** B, C and E only
- **(2)** C, D and E only
- **(3)** A, B and D only
- **(4)** A, B and E only

Q161. Match (Enzyme) ↔ (Bond cleaved): A. Lipase B. Nuclease C. Protease D. Amylase ↔ I. Peptide bond II. Ester bond III. Glycosidic bond IV. Phosphodiester bond

- **(1)** A-II, B-IV, C-I, D-III
- **(2)** A-IV, B-I, C-III, D-II
- **(3)** A-IV, B-II, C-III, D-I
- **(4)** A-III, B-II, C-I, D-IV

Q162. Flippers of penguins and dolphins are an example of:

- **(1)** Convergent evolution
- **(2)** Divergent evolution
- **(3)** Adaptive radiation
- **(4)** Natural selection

Q163. Match: A. Expiratory capacity B. Functional residual capacity C. Vital capacity D. Inspiratory capacity ↔ I. ERV+TV+IRV II. TV+ERV III. TV+IRV IV. ERV+RV

- **(1)** A-II, B-I, C-IV, D-III
- **(2)** A-I, B-III, C-II, D-IV
- **(3)** A-II, B-IV, C-I, D-III
- **(4)** A-III, B-II, C-IV, D-I

Q164. Which factor will NOT affect Hardy-Weinberg equilibrium?

- **(1)** Gene migration
- **(2)** Constant gene pool
- **(3)** Genetic recombination
- **(4)** Genetic drift

Q165. Sequence of human evolution (past to recent): A. Homo habilis B. Homo sapiens C. Homo neanderthalensis D. Homo erectus

- **(1)** C-B-D-A
- **(2)** A-D-C-B
- **(3)** D-A-C-B
- **(4)** B-A-D-C

Q166. Pathway of action potential through heart: A. AV bundle B. Purkinje fibres C. AV node D. Bundle branches E. SA node

- **(1)** B-D-E-C-A
- **(2)** E-A-D-B-C
- **(3)** E-C-A-D-B
- **(4)** A-E-C-B-D

Q167. Factors favouring oxyhaemoglobin formation in alveoli:

- **(1)** Low pCO■ and high H■
- **(2)** Low pCO■ and high temperature
- **(3)** High pO■ and high pCO■
- **(4)** High pO■ and lesser H■ concentration

Q168. Match: A. α-1 antitrypsin B. Cry IAb C. Cry IAc D. Enzyme replacement therapy $\leftrightarrow$ I. Cotton bollworm II. ADA deficiency III. Emphysema IV. Corn borer

- **(1)** A-III, B-IV, C-I, D-II
- **(2)** A-II, B-IV, C-I, D-III
- **(3)** A-II, B-I, C-IV, D-III
- **(4)** A-III, B-I, C-II, D-IV

Q169. Assertion: FSH acts on ovarian follicles in females and Leydig cells in males. Reason: Growing ovarian follicles secrete estrogen; interstitial cells secrete androgens.

- **(1)** A true, R false
- **(2)** A false, R true
- **(3)** Both true; R is correct explanation
- **(4)** Both true; R is NOT correct explanation

Q170. In E. coli cloning vector pBR322, the genes 'X' and 'Y' represent:

- **(1)** X: replication protein; Y: antibiotic resistance
- **(2)** X: recognition sites; Y: antibiotic resistance
- **(3)** X: antibiotic resistance; Y: replication protein
- **(4)** X: copy-number control; Y: replication protein

Q171. Match: A. Cocaine B. Heroin C. Morphine D. Marijuana $\leftrightarrow$ I. Effective sedative in surgery II. Cannabis sativa III. Erythroxylum IV. Papaver somniferum

- **(1)** A-II, B-I, C-III, D-IV
- **(2)** A-III, B-IV, C-I, D-II
- **(3)** A-IV, B-III, C-I, D-II
- **(4)** A-I, B-III, C-II, D-IV

Q172. Consider: A. Annelids are true coelomates B. Poriferans are pseudocoelomates C. Aschelminthes are acoelomates D. Platyhelminthes are pseudocoelomates

- **(1)** C only
- **(2)** D only
- **(3)** B only
- **(4)** A only

Q173. Statement I: In the nephron, the descending limb of loop of Henle is impermeable to water and permeable to electrolytes. Statement II: The PCT is lined by simple columnar brush border epithelium and increases surface area for reabsorption.

(1) I true, II false
(2) I false, II true
(3) Both true
(4) Both false

Q174. Match: A. Fibrous joints B. Cartilaginous joints C. Hinge D. Ball-and-socket ↔ I. Adjacent vertebrae, limited movement II. Humerus and pectoral girdle, rotational III. Skull, no movement IV. Knee, locomotion

(1) A-II, B-III, C-I, D-IV
(2) A-III, B-I, C-IV, D-II
(3) A-IV, B-II, C-III, D-I
(4) A-I, B-III, C-II, D-IV

Q175. Which is NOT a natural/traditional contraceptive method?

(1) Lactational amenorrhea
(2) Vaults
(3) Coitus interruptus
(4) Periodic abstinence

Q176. Match: A. Pleurobrachia B. Radula C. Stomochord D. Air bladder ↔ I. Mollusca II. Ctenophora III. Osteichthyes IV. Hemichordata

(1) A-II, B-IV, C-I, D-III
(2) A-IV, B-III, C-II, D-I
(3) A-IV, B-II, C-III, D-I
(4) A-II, B-I, C-IV, D-III

Q177. Match: A. Axoneme B. Cartwheel pattern C. Crista D. Satellite ↔ I. Centriole II. Cilia and flagella III. Chromosome IV. Mitochondria

(1) A-II, B-IV, C-I, D-III
(2) A-II, B-I, C-IV, D-III
(3) A-IV, B-III, C-II, D-I
(4) A-IV, B-II, C-III, D-II

Q178. Which statement is INCORRECT?

(1) Bioreactors are used to produce small-scale bacterial cultures
(2) Bioreactors have agitator, oxygen-delivery, and foam-control systems
(3) A bioreactor provides optimal growth conditions for the desired product
(4) Most commonly used bioreactors are stirring-type

Q179. Match (Sub-phases of Prophase-I) with (Specific characters): A. Diakinesis B. Pachytene C. Zygotene D. Leptotene ↔ I. Synaptonemal complex formation II. Completion of terminalisation of chiasmata III. Chromosomes look like thin threads IV. Appearance of recombination nodules

(1) A-II, B-IV, C-I, D-III
(2) A-IV, B-III, C-II, D-I
(3) A-IV, B-II, C-III, D-I
(4) A-I, B-II, C-IV, D-III

Q180. Match: A. Common cold B. Haemozoin C. Widal test D. Allergy ↔ I. Plasmodium II. Typhoid III. Rhinoviruses IV. Dust mites

- **(1)** A-III, B-I, C-II, D-IV
- **(2)** A-IV, B-II, C-III, D-I
- **(3)** A-II, B-IV, C-III, D-I
- **(4)** A-I, B-III, C-II, D-IV

Q181. Assertion: Breastfeeding during initial period of infant growth is recommended. Reason: Colostrum contains several antibodies essential to develop resistance for the new born baby.

- **(1)** A correct, R not correct
- **(2)** A not correct, R correct
- **(3)** Both correct; R correct explanation
- **(4)** Both correct; R NOT correct explanation

Q182. Match: A. Pterophyllum B. Myxine C. Pristis D. Exocoetus ↔ I. Hag fish II. Saw fish III. Angel fish IV. Flying fish

- **(1)** A-IV, B-I, C-II, D-III
- **(2)** A-III, B-II, C-I, D-IV
- **(3)** A-II, B-I, C-III, D-IV
- **(4)** A-III, B-I, C-II, D-IV

Q183. 'Ti plasmid' of Agrobacterium tumefaciens stands for:

- **(1)** Tumour-inducing plasmid
- **(2)** Temperature-independent plasmid
- **(3)** Tumour-inhibiting plasmid
- **(4)** Tumour-independent plasmid

Q184. Which is NOT a component of the Fallopian tube?

- **(1)** Infundibulum
- **(2)** Ampulla
- **(3)** Uterine fundus
- **(4)** Isthmus

Q185. Match: A. Down's syndrome B. α-Thalassemia C. β-Thalassemia D. Klinefelter's ↔ I. 11th chromosome II. 'X' chromosome III. 21st chromosome IV. 16th chromosome

- **(1)** A-III, B-IV, C-I, D-II
- **(2)** A-IV, B-I, C-II, D-III
- **(3)** A-I, B-II, C-III, D-IV
- **(4)** A-II, B-III, C-IV, D-I

SECTION B — Q36 to Q50 of the subject (attempt any 10)

Q186. Statements about non-chordates: A. Pharynx perforated by gill slits B. Notochord absent C. CNS dorsal D. Heart dorsal if present E. Post-anal tail absent

- **(1)** B, D and E only
- **(2)** B, C and D only
- **(3)** A and C only
- **(4)** A, B and D only

Q187. Match (Era) ↔ (Group): A. Mesozoic B. Proterozoic C. Cenozoic D. Paleozoic ↔ I. Lower invertebrates II. Fish & Amphibia III. Birds & Reptiles IV. Mammals

- **(1)** A-I, B-II, C-IV, D-III
- **(2)** A-III, B-I, C-IV, D-II
- **(3)** A-II, B-I, C-III, D-IV
- **(4)** A-III, B-I, C-II, D-IV

Q188. Statement I: Cerebral hemispheres are connected by corpus callosum. Statement II: Brain stem consists of medulla oblongata, pons, and cerebrum.

- **(1)** I correct, II incorrect
- **(2)** I incorrect, II correct
- **(3)** Both correct
- **(4)** Both incorrect

Q189. Spermatogenesis pathway: GnRH → LH/A → B/C → Androgens/Factors → spermatids/D. Identify A, B, C, D:

- **(1)** FSH, Sertoli cells, Leydig cells, spermatogenesis
- **(2)** ICSH, Leydig cells, Sertoli cells, spermatogenesis
- **(3)** FSH, Leydig cells, Sertoli cells, spermiogenesis
- **(4)** ICSH, Interstitial cells, Leydig cells, spermiogenesis

Q190. Match: A. RNA polymerase III B. Termination of transcription C. Splicing of exons D. TATA box ↔ I. snRNPs II. Promoter III. Rho factor IV. snRNAs, tRNA

- **(1)** A-III, B-IV, C-I, D-II
- **(2)** A-IV, B-III, C-I, D-II
- **(3)** A-II, B-IV, C-I, D-III
- **(4)** A-III, B-II, C-IV, D-I

Q191. Match: A. Exophthalmic goitre B. Acromegaly C. Cushing's syndrome D. Cretinism ↔ I. ↑cortisol, moon face, hyperglycaemia II. ↓thyroid, stunted growth III. ↑thyroid + protruding eyeballs IV. ↑Growth Hormone

- **(1)** A-III, B-IV, C-II, D-I
- **(2)** A-III, B-IV, C-I, D-II
- **(3)** A-I, B-III, C-II, D-IV
- **(4)** A-IV, B-II, C-I, D-III

Q192. Match: A. Unicellular glandular epithelium B. Compound epithelium C. Multicellular glandular epithelium D. Endocrine glandular epithelium ↔ I. Salivary glands II. Pancreas III. Goblet cells IV. Buccal cavity moist surface

- **(1)** A-III, B-IV, C-I, D-II
- **(2)** A-II, B-I, C-IV, D-III
- **(3)** A-II, B-I, C-III, D-IV
- **(4)** A-IV, B-III, C-I, D-II

Q193. Statement I: Bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced. Statement II: Both bone marrow and thymus provide microenvironments for development and maturation of T-lymphocytes.

- **(1)** I correct, II incorrect
- **(2)** I incorrect, II correct
- **(3)** Both correct
- **(4)** Both incorrect

Q194. Match: A. Storing food B. Ring of 6-8 blind tubules at junction of foregut/midgut C. Ring of 100-150 yellow filaments at midgut/hindgut D. Grinding food ↔ I. Gizzard II. Gastric Caeca III. Malpighian tubules IV. Crop

- **(1)** A-IV, B-III, C-II, D-I
- **(2)** A-III, B-II, C-IV, D-I
- **(3)** A-IV, B-II, C-III, D-I
- **(4)** A-I, B-II, C-III, D-IV

Q195. Correct statement on juxtamedullary nephron:

- **(1)** Loop of Henle runs deep into medulla
- **(2)** Outnumber cortical nephrons
- **(3)** Located in columns of Bertini
- **(4)** Renal corpuscle lies in outer renal medulla

Q196. Match (ECG): A. P wave B. QRS complex C. T wave D. T-P gap ↔ I. Heart muscles electrically silent II. Depolarisation of ventricles III. Depolarisation of atria IV. Repolarisation of ventricles

- **(1)** A-II, B-III, C-I, D-IV
- **(2)** A-IV, B-II, C-I, D-III
- **(3)** A-I, B-III, C-IV, D-II
- **(4)** A-III, B-II, C-IV, D-I

Q197. Father B■, mother A■, child O■. Possible genotypes: A. I■I/I■I/ii B. I■I■/I■I■/ii C. I■I■/iI■/I■i D. I■I/I■I/I■I E. iI■/iI■/I■I■

- **(1)** C and B only
- **(2)** D and E only
- **(3)** A only
- **(4)** B only

Q198. Statement I: Gause's competitive exclusion principle states two closely related species competing for different resources cannot coexist. Statement II: According to Gause's principle, in competition the inferior is eliminated, true if resources are limiting.

- **(1)** I true, II false
- **(2)** I false, II true
- **(3)** Both true
- **(4)** Both false

Q199. Catalytic cycle of enzyme action — correct sequence: A. ES complex formation B. Free enzyme ready C. Release of products D. Bonds broken E. Substrate binding

- **(1)** B, A, C, D, E
- **(2)** E, D, C, B, A
- **(3)** E, A, D, C, B
- **(4)** A, E, B, D, C

Q200. Statement I: Mitochondria and chloroplasts are both double-membrane bound. Statement II: Inner membrane of mitochondria is relatively less permeable than chloroplast.

- **(1)** I correct, II incorrect
- **(2)** I incorrect, II correct
- **(3)** Both correct
- **(4)** Both incorrect

Official Answer Key — NEET (UG) 2024

All answers are referenced to **Test Booklet Code T3** as released by the conducting authority.

Q. No.	Ans	Q. No.	Ans	Q. No.	Ans	Q. No.	Ans
1	(4)	51	(1)	101	(1)	151	(4)
2	(1)	52	(4)	102	(4)	152	(2)
3	(3)	53	(4)	103	(4)	153	(1)
4	(2)	54	(3)	104	(3)	154	(1)
5	(4)	55	(1)	105	(3)	155	(2)
6	(2)	56	(1)	106	(2)	156	(1)
7	(3)	57	(3)	107	(1)	157	(1)
8	(3)	58	(3)	108	(1)	158	(3)
9	(3)	59	(2)	109	(1)	159	(3)
10	(3)	60	(4)	110	(1)	160	(4)
11	(2)	61	(3)	111	(3)	161	(4)
12	(4)	62	(2)	112	(1)	162	(4)
13	(3)	63	(2)	113	(3)	163	(4)
14	(4)	64	(2)	114	(4)	164	(2)
15	(2)	65	(3)	115	(2)	165	(1)
16	(1)	66	(3)	116	(1)	166	(1)
17	(1)	67	(3)	117	(1)	167	(1)
18	(4)	68	(3)	118	(4)	168	(3)
19	(3)	69	(3)	119	(3)	169	(3)
20	(3)	70	(4)	120	(1)	170	(4)
21	(1)	71	(4)	121	(2)	171	(2)
22	(2)	72	(2)	122	(2)	172	(4)
23	(1)	73	(3)	123	(3)	173	(1)
24	(4)	74	(1)	124	(2)	174	(3)
25	(4)	75	(3)	125	(1)	175	(4)
26	(3)	76	(3)	126	(3)	176	(2)
27	(4)	77	(2)	127	(4)	177	(1)
28	(1)	78	(1)	128	(2)	178	(2)
29	(2)	79	(3)	129	(3)	179	(2)
30	(1)	80	(2)	130	(4)	180	(2)
31	(4)	81	(1)	131	(3)	181	(4)
32	(2)	82	(3)	132	(2)	182	(2)
33	(2)	83	(4)	133	(1)	183	(2)
34	(4)	84	(2)	134	(4)	184	(4)
35	(3)	85	(3)	135	(4)	185	(4)
36	(3)	86	(3)	136	(4)	186	(3)

37	(4)	87	(2)	137	(4)	187	(4)
38	(2)	88	(2)	138	(3)	188	(1)
39	(4)	89	(4)	139	(3)	189	(1)
40	(4)	90	(3)	140	(2)	190	(3)
41	(4)	91	(3)	141	(4)	191	(2)
42	(4)	92	(3)	142	(1)	192	1 or 3
43	(4)	93	(3)	143	(4)	193	(3)
44	(4)	94	(2)	144	(4)	194	(2)
45	(3)	95	(4)	145	(1)	195	(2)
46	(4)	96	(4)	146	(1)	196	(3)
47	(4)	97	(3)	147	(1)	197	(1)
48	(1)	98	(2)	148	1 or 4	198	(1)
49	(2)	99	(4)	149	(3)	199	(2)
50	(4)	100	(3)	150	(3)	200	(3)

Marking Scheme: Correct: +4 | Incorrect: –1 | Unattempted: 0

Note: For Q148 and Q192, multiple options were considered correct as per the official key.